

PAPER_675: UQFF Analysis of GW170817: NS-NS Merger, GRB Delay, and Tidal Deformability

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Framework: Unified Quantum Field Framework (UQFF) v5.30

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Class: UQFFComparedToGW170817 | CP4 #259

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Domain: Multi-Messenger Astrophysics

Abstract

GW170817, the first binary neutron star merger detected by LIGO/Virgo (August 17, 2017), provides the most stringent multi-messenger test of modified gravity. The UQFF predicts a modified GRB delay $\Delta t_{\text{UQFF}} = 1.7 (1 + f_{\text{TRZ}} \rho_{\text{UA}} / \rho_{\text{SCm}})$ s and a reduced tidal deformability $\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}} (1 - f_{\text{TRZ}} \rho_{\text{SCm}} / \rho_{\text{UA}})$. Both effects are consistent with the observed 1.74 s EM/GW delay within UQFF parameter uncertainties.

Primary UQFF Equation

$$\Delta t_{\text{UQFF}} = 1.7 \left(1 + f_{\text{TRZ}} \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} \right) \text{ s}$$

Parameters

Parameter	Value	Description
m_1	$1.36 M_{\odot}$	NS1 mass
m_2	$1.17 M_{\odot}$	NS2 mass
d	40 Mpc	Distance
f_{peak}	1500 Hz	Post-merger freq

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate

Constant	Value	Description
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "UQFFComparedToGW170817.h"
```

```
UQFFComparedToGW170817 obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFComparedToGW170817Calculator

calc = UQFFComparedToGW170817Calculator()
result = calc.compute() # returns all UQFF quantities
print(result)
```

References

Abbott et al. 2017 (GW170817); Monitor of All-sky X-ray Image (MAXI); UQFF Session 173

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